CO2-AT1-SCENARIOBASED QUESTIONS

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SUB CODE:CSA0927 **REG no.:192425201**

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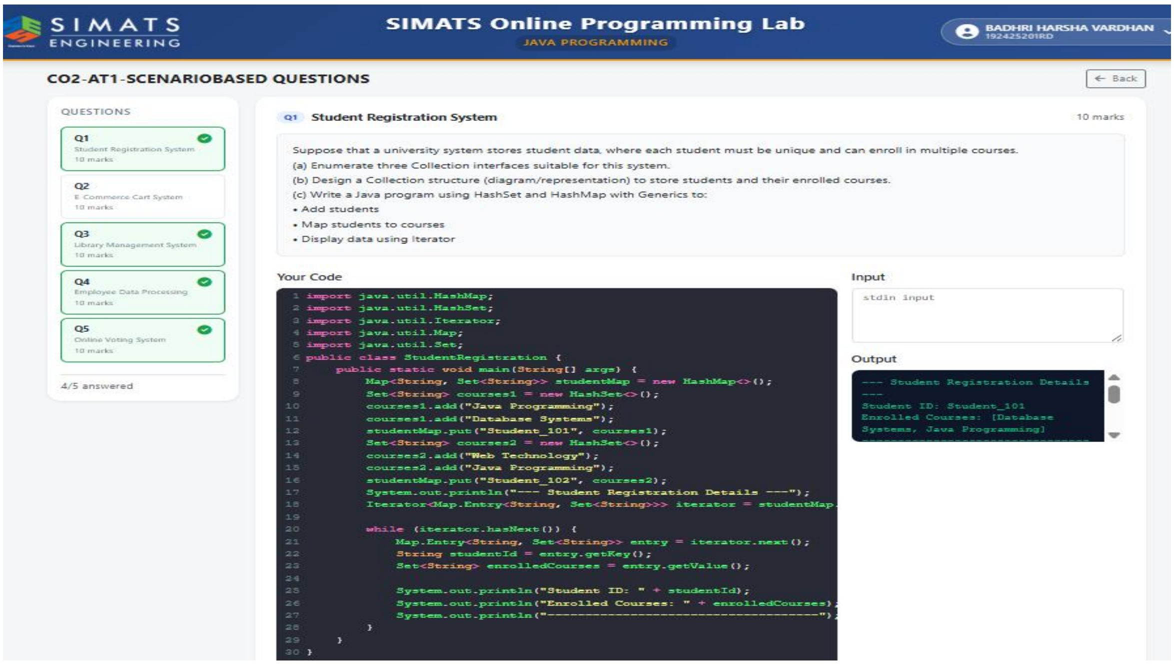

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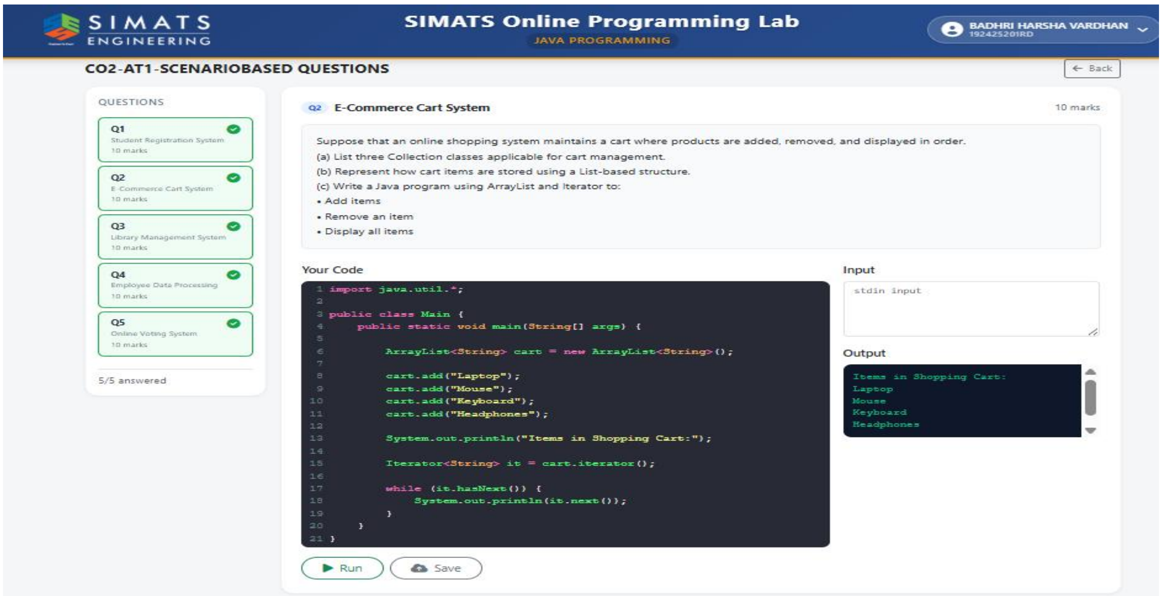

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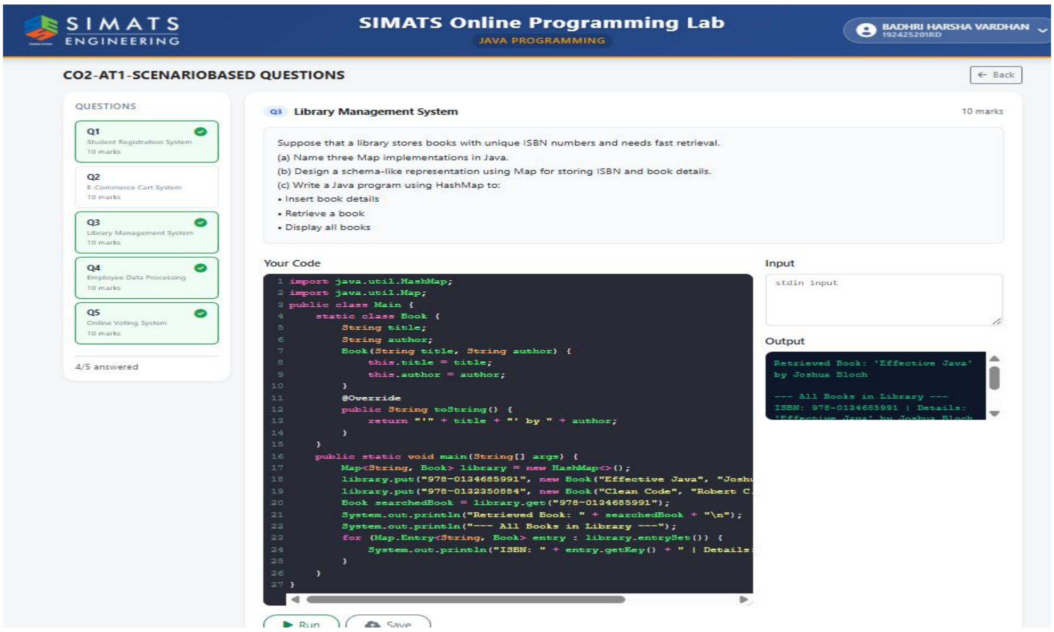

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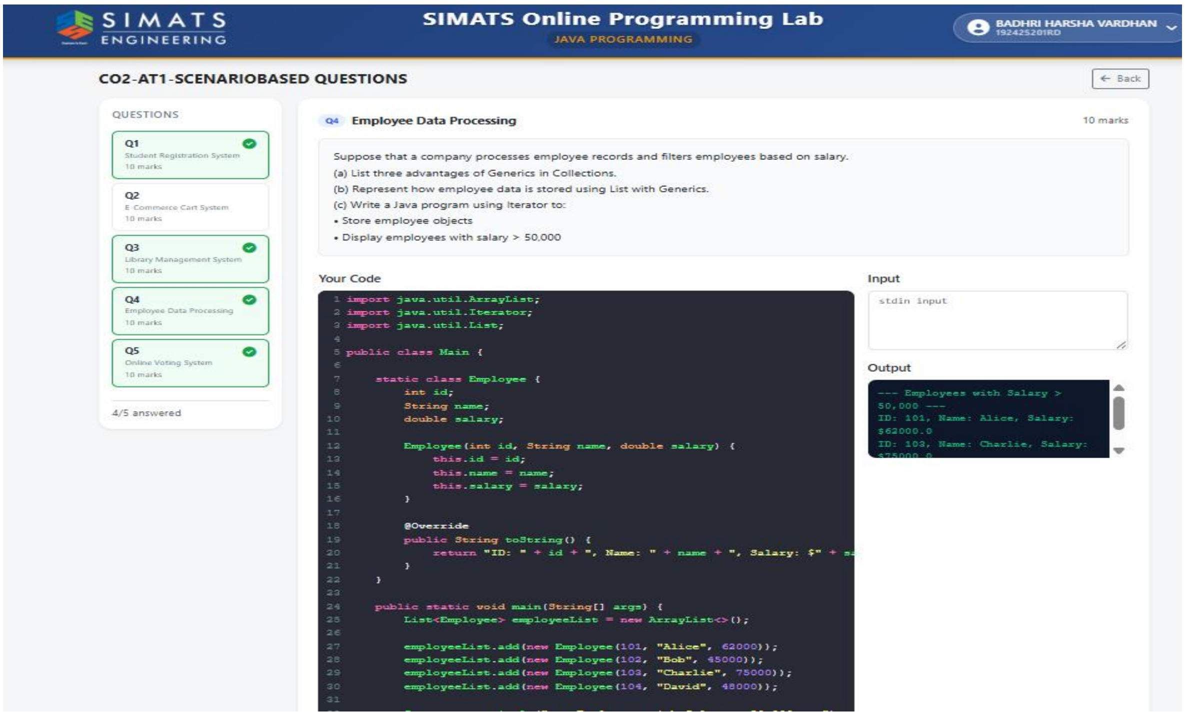
CO2-AT1-SCENARIOBASED QUESTIONS
Back
QUESTIONS
Q1
Student Registration System
10 marks
Q2
E Commerce Cart System
10 marks
Q3
Library Management System
10 marks
Q4
Employee Data Processing
10 marks
Q5
Online Voting System
10 marks
4/5 answered
Q3 Library Management System
10 marks
Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:
• Insert book details
• Retrieve a book
• Display all books
Your Code
import java.util.HashMap;
import java.util.Map;
public class Main {
static class Book {
String title;
String author;
Book(String title, String author) {
this.title = title;
this.author = author;
}
@Override
public String toString() {
return "'" + title + "' by " + author;
}
}
public static void main(String[] args) {
Map<String, Book> library = new HashMap<>();
library.put("978-0134685991", new Book("Effective Java", "Joshu
library.put("978-0132350884", new Book("Clean Code", "Robert C.
Book searchedBook = library.get("978-0134685991");
System.out.println("Retrieved Book: " + searchedBook + "\n");
System.out.println("--- All Books in Library ---");
for (Map.Entry<String, Book> entry : library.entrySet()) {
System.out.println("ISBN: " + entry.getKey() + " | Details:
}
}
}
Input
stdin input
Output
Retrieved Book: 'Effective Java' by Joshua Bloch
--- All Books in Library ---
ISBN: 978-0134685991 | Details:
Run
Save

4.



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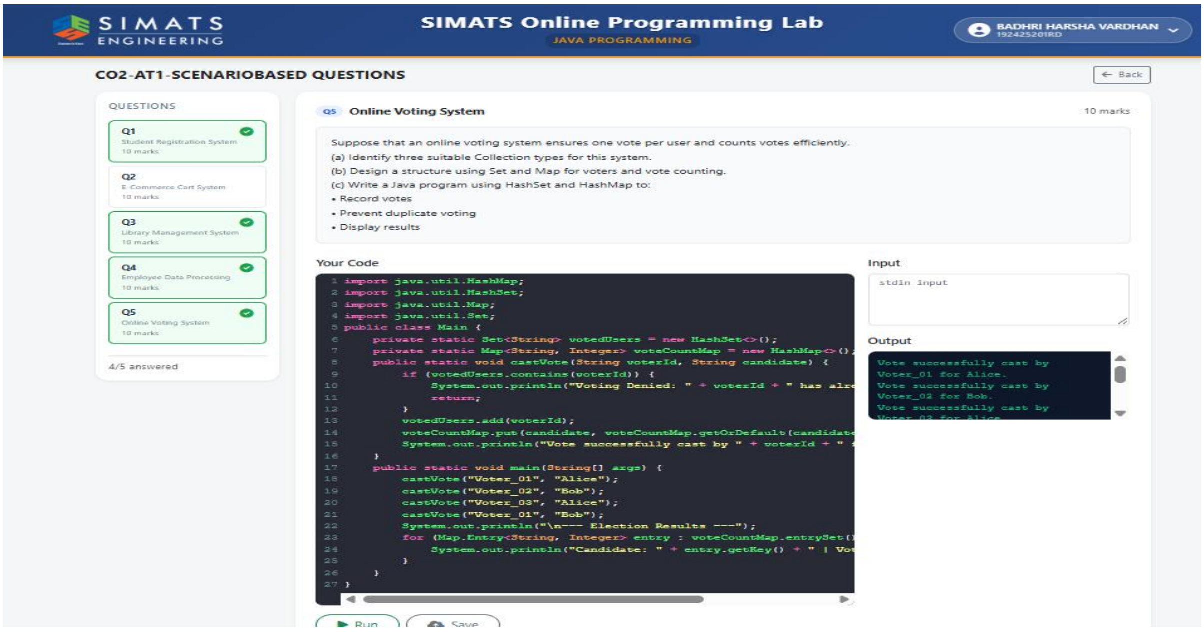
10 marks
4/5 answered
Q4 Employee Data Processing
10 marks
Suppose that a company processes employee records and filters employees based on salary.
(a) List three advantages of Generics in Collections.
(b) Represent how employee data is stored using List with Generics.
(c) Write a Java program using Iterator to:
• Store employee objects
• Display employees with salary > 50,000
Your Code
import java.util.ArrayList;
import java.util.Iterator;
import java.util.List;
public class Main {
static class Employee {
int id;
String name;
double salary;
Employee(int id, String name, double salary) {
this.id = id;
this.name = name;
this.salary = salary;
}
@Override
public String toString() {
return "ID: " + id + ", Name: " + name + ", Salary: $" + sa
}
}
public static void main(String[] args) {
List<Employee> employeeList = new ArrayList<>();
employeeList.add(new Employee(101, "Alice", 62000));
employeeList.add(new Employee(102, "Bob", 45000));
employeeList.add(new Employee(103, "Charlie", 75000));
employeeList.add(new Employee(104, "David", 48000));
Input
stdin input
Output
--- Employees with Salary > 50,000 ---
ID: 101, Name: Alice, Salary: $62000.0
ID: 103, Name: Charlie, Salary:

5.



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Online Voting System
10 marks
4/5 answered
Q5 Online Voting System
10 marks
Suppose that an online voting system ensures one vote per user and counts votes efficiently.
(a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
• Record votes
• Prevent duplicate voting
• Display results
Your Code
import java.util.HashMap;
import java.util.HashSet;
import java.util.Map;
import java.util.Set;
public class Main {
private static Set<String> votedUsers = new HashSet<>();
private static Map<String, Integer> voteCountMap = new HashMap<>();
public static void castVote(String voterId, String candidate) {
if (votedUsers.contains(voterId)) {
System.out.println("Voting Denied: " + voterId + " has alre
return;
}
votedUsers.add(voterId);
voteCountMap.put(candidate, voteCountMap.getOrDefault(candidate
System.out.println("Vote successfully cast by " + voterId + "
}
public static void main(String[] args) {
castVote("Voter_01", "Alice");
castVote("Voter_02", "Bob");
castVote("Voter_03", "Alice");
castVote("Voter_01", "Bob");
System.out.println("\n--- Election Results ---");
for (Map.Entry<String, Integer> entry : voteCountMap.entrySet()
System.out.println("Candidate: " + entry.getKey() + " | Vot
}
}
}
Input
stdin input
Output
Vote successfully cast by
Voter_01 for Alice.
Vote successfully cast by
Voter_02 for Bob.
Vote successfully cast by
Run
Save